北京大学
PEKING UNIVERSITY

第二章 原子结构与光谱

3. 原子光谱

跃迁（transition）

允许跃迁与禁阻跃迁

选择定则

跃迁过程**角动量守恒**

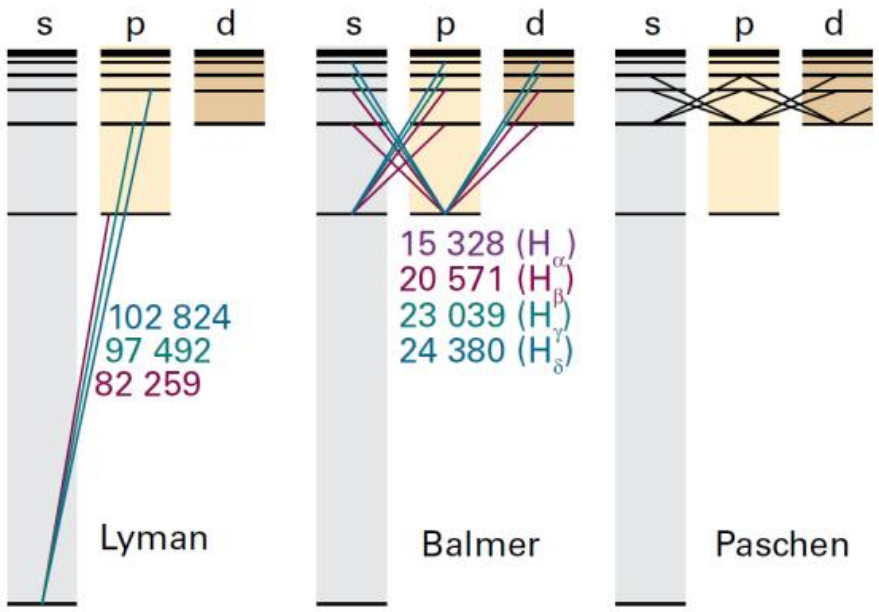
跃迁过程**吸收或释放光子(s=1)**

从而轨道角动量必须发生改变，
且改变量$\Delta l = \pm 1$

$$S = \hbar\sqrt{s(s+1)}$$

电磁场量子化 → 量子电动力学

本章电磁场按经典力学处理



偶极近似(dipole approximation)

跃迁偶极矩(transition dipole moment)

$$\mu_{fi} = \int \psi_f^* \hat{\mu} \psi_i \, d\tau$$

允许跃迁对应非零跃迁偶极矩

问题一　微扰理论

微扰理论（perturbation theory）包括定态微扰与含时微扰

定态微扰

近似求解薛定谔方程

$$\hat{H}\psi_n = E_n\psi_n \qquad \hat{H}=\hat{H}^{(0)}+\hat{H}^{(1)}$$

已知

$$\hat{H}^{(0)}\psi_n^{(0)} = E_n^{(0)}\psi_n^{(0)}$$

令λ = 1， 得

$$E=E^{(0)}+E^{(1)}+E^{(2)}+\ldots$$

$$\psi=\psi^{(0)}+\psi^{(1)}+\psi^{(2)}+\ldots$$

以基态为例，考虑微扰展开 $\hat{H}=\hat{H}^{(0)}+\lambda\hat{H}^{(1)}$

$$E_0=E_0^{(0)}+\lambda E_0^{(1)}+\lambda^2E_0^{(2)}+\cdots \qquad \psi_0=\psi_0^{(0)}+\lambda\psi_0^{(1)}+\lambda^2\psi_0^{(2)}+\cdots$$

代入薛定谔方程，比较各阶小量

$$(\hat{H}^{(0)}+\lambda\hat{H}^{(1)})(\psi_0^{(0)}+\lambda\psi_0^{(1)}+\lambda^2\psi_0^{(2)}+\cdots)=$$
$$(E_0^{(0)}+\lambda E_0^{(1)}+\lambda^2E_0^{(2)}+\cdots)(\psi_0^{(0)}+\lambda\psi_0^{(1)}+\lambda^2\psi_0^{(2)}+\cdots)$$

$$\hat{H}^{(0)}\psi_0^{(0)}+\lambda(\hat{H}^{(1)}\psi_0^{(0)}+\hat{H}^{(1)}\psi_0^{(1)})+\lambda^2(\hat{H}^{(0)}\psi_0^{(2)}+\hat{H}^{(1)}\psi_0^{(1)})+\cdots$$
$$=E_0^{(0)}\psi_0^{(0)}+\lambda(E_0^{(0)}\psi_0^{(1)}+E_0^{(1)}\psi_0^{(0)})+\lambda^2(E_0^{(2)}\psi_0^{(0)}$$
$$+E_0^{(1)}\psi_0^{(1)}+E_0^{(0)}\psi_0^{(2)})+\cdots$$

Terms in λ^0: $\hat{H}^{(0)}\psi_0^{(0)} = E_0^{(0)}\psi_0^{(0)}$ 能量零阶，自动满足

Terms in λ^1: $\hat{H}^{(1)}\psi_0^{(0)} + \hat{H}^{(0)}\psi_0^{(1)} = E_0^{(0)}\psi_0^{(1)} + E_0^{(1)}\psi_0^{(0)}$ 求能量一阶修正

Terms in λ^2: $\hat{H}^{(0)}\psi_0^{(2)} + \hat{H}^{(1)}\psi_0^{(1)} = E_0^{(2)}\psi_0^{(0)} + E_0^{(1)}\psi_0^{(1)} + E_0^{(0)}\psi_0^{(2)}$ 求能量二阶修正

一阶修正方程与$\psi_0^{(0)}$做内积，化为

$$\langle\psi_0^{(0)}|\hat{H}^{(1)}|\psi_0^{(0)}\rangle + \langle\psi_0^{(0)}|\hat{H}^{(0)}|\psi_0^{(1)}\rangle = E_0^{(0)}\langle\psi_0^{(0)}|\psi_0^{(1)}\rangle + E_0^{(1)}\langle\psi_0^{(0)}|\psi_0^{(0)}\rangle$$

从而**能量一阶修正** $E_0^{(1)} = \langle\psi_0^{(0)}|\hat{H}^{(1)}|\psi_0^{(0)}\rangle$

一阶修正方程与$\psi_k^{(0)}$做内积，得到

$$\langle\psi_k^{(0)}|\hat{H}^{(1)}|\psi_0^{(0)}\rangle + \langle\psi_k^{(0)}|\hat{H}^{(0)}|\psi_0^{(1)}\rangle = E_0^{(0)}\langle\psi_k^{(0)}|\psi_0^{(1)}\rangle + E_0^{(1)}\langle\psi_k^{(0)}|\psi_0^{(0)}\rangle, \quad k \neq 0$$

$$\langle\psi_k^{(0)}|\hat{H}^{(0)}|\psi_0^{(1)}\rangle = E_k^{(0)}\langle\psi_k^{(0)}|\psi_0^{(1)}\rangle \quad \Longrightarrow \quad \langle\psi_k^{(0)}|\psi_0^{(1)}\rangle = \frac{\langle\psi_k^{(0)}|\hat{H}^{(1)}|\psi_0^{(0)}\rangle}{E_0^{(0)} - E_k^{(0)}}$$

波函数一阶修正 $|\psi_0^{(1)}\rangle = \sum_k |\psi_k^{(0)}\rangle\langle\psi_k^{(0)}|\psi_0^{(1)}\rangle$ $\psi_0^{(1)} = \sum_{k\neq 0} \frac{\langle\psi_k^{(0)}|\hat{H}^{(1)}|\psi_0^{(0)}\rangle}{E_0^{(0)} - E_k^{(0)}}\psi_k^{(0)}$

一阶修正不含$\psi_0^{(0)}$
(见下页)

说明：ψ_λ的归一化条件决定$\psi_0^{(1)}$与$\psi_0^{(0)}$正交！

$$\langle\psi_\lambda|\psi_\lambda\rangle = 1 \quad\Longrightarrow\quad \langle\psi_0^{(0)} + \lambda\psi_0^{(1)} + \cdots|\psi_0^{(0)} + \lambda\psi_0^{(1)} + \cdots\rangle = 1$$

$$\Longrightarrow\quad \langle\psi_0^{(0)}|\psi_0^{(1)}\rangle = 0$$

假定实值波函数

Terms in λ^2: $\hat{H}^{(0)}\psi_0^{(2)} + \hat{H}^{(1)}\psi_0^{(1)} = E_0^{(2)}\psi_0^{(0)} + E_0^{(1)}\psi_0^{(1)} + E_0^{(0)}\psi_0^{(2)}$ 求能量二阶修正

二阶修正方程与$\psi_0^{(0)}$做内积，化为

$$\langle\psi_0^{(0)}|\hat{H}^{(0)}|\psi_0^{(2)}\rangle + \langle\psi_0^{(0)}|\hat{H}^{(1)}|\psi_0^{(1)}\rangle = E_0^{(2)}\langle\psi_0^{(0)}|\psi_0^{(0)}\rangle + E_0^{(1)}\langle\psi_0^{(0)}|\psi_0^{(1)}\rangle + E_0^{(0)}\langle\psi_0^{(0)}|\psi_0^{(2)}\rangle$$

$$\langle\psi_0^{(0)}|\psi_0^{(1)}\rangle = 0$$

从而

$$E_0^{(2)} = \langle\psi_0^{(0)}|\hat{H}^{(1)}|\psi_0^{(1)}\rangle$$

代入一阶波函数修正，得**二阶能量修正**

$$\psi_0^{(1)} = \sum_{k\neq 0}\frac{\langle\psi_k^{(0)}|\hat{H}^{(1)}|\psi_0^{(0)}\rangle}{E_0^{(0)} - E_k^{(0)}}\psi_k^{(0)}$$

$$E_0^{(2)} = \sum_{k\neq 0}\frac{|\langle\psi_k^{(0)}|\hat{H}^{(1)}|\psi_0^{(0)}\rangle|^2}{E_0^{(0)} - E_k^{(0)}}$$

物理解读：

- 波函数不弛豫，得能量一阶修正
- 考虑波函数一阶弛豫，得能量二阶修正
- 能量二阶修正恒为负

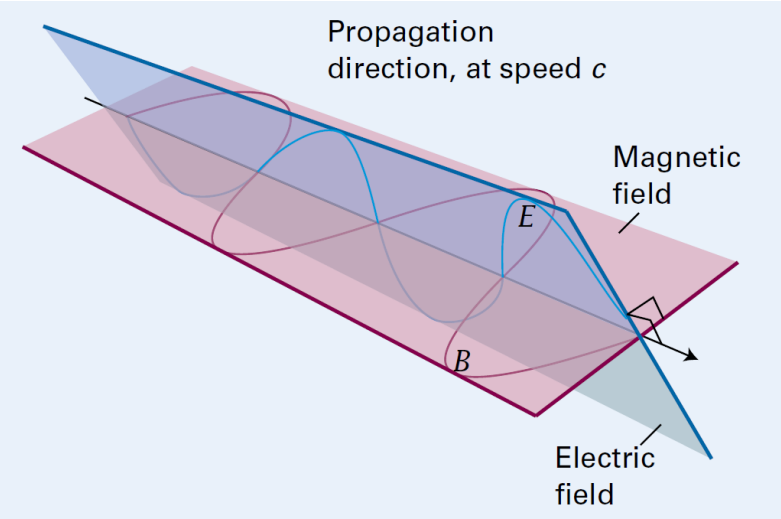
含时微扰

含时薛定谔方程 $i\hbar\partial_t\Psi(\boldsymbol{r},t)=\hat{H}\Psi(\boldsymbol{r},t)$

假如H不含时，且$\psi(\boldsymbol{r})$是H的定态 $\hat{H}\psi(\boldsymbol{r})=E\psi(\boldsymbol{r})$

含时波函数按照下式演化 $\Psi(\boldsymbol{r},t)=\psi(\boldsymbol{r})e^{-\frac{iEt}{\hbar}}$

可观测量不随时间变化 $\langle\Psi(\boldsymbol{r},t)|\hat{\Omega}|\Psi(\boldsymbol{r},t)\rangle=\langle\psi(\boldsymbol{r})|\hat{\Omega}|\psi(\boldsymbol{r})\rangle$



定态(stationary state)的来由
前提是算符不含时

含时问题，将**含时部分视为微扰**

$$\hat{H}=\hat{H}^{(0)}+\hat{H}^{(1)}(t)$$

例如原子分子与电磁场的相互作用

$$\hat{H}^{(1)}(t)=-\hat{\mu}_z\mathcal{E}\cos\omega t$$

偶极近似下，电磁场处理为局部匀强电场

z为电场振荡方向，μ_z为偶极算符

考虑一阶微扰波函数

$$|\Psi_0^{(1)}(t)\rangle=\sum_{k\neq0}a_k(t)|\Psi_k^{(0)}(t)\rangle$$

推导$a_k(t)$表达式

定态微扰:

$$\hat{H} = \hat{H}^{(0)} + \lambda \hat{H}^{(1)}$$

$$|\psi\rangle = |\psi_0^{(0)}\rangle + \lambda|\psi_0^{(1)}\rangle + \lambda^2|\psi_0^{(2)}\rangle + \cdots$$

$$\langle\psi_k^{(0)}|E_0^{(0)} - \hat{H}^{(0)}|\psi_0^{(1)}\rangle = \langle\psi_k^{(0)}|\hat{H}^{(1)}|\psi_0^{(0)}\rangle$$

$$(E_k^{(0)} - \hat{H}^{(0)})\psi_k^{(0)} = 0$$

$$|\psi_0^{(1)}\rangle = \sum_{k\neq 0} c_k|\psi_k^{(0)}\rangle$$

$$c_k = \frac{\langle\psi_k^{(0)}|\hat{H}^{(1)}|\psi_0^{(0)}\rangle}{E_0^{(0)} - E_k^{(0)}}$$

含时微扰:

$$\hat{H}(t) = \hat{H}^{(0)} + \lambda \hat{H}^{(1)}(t)$$

$$|\Psi(t)\rangle = |\Psi_0^{(0)}(t)\rangle + \lambda|\Psi_0^{(1)}(t)\rangle + \lambda^2|\Psi_0^{(2)}(t)\rangle + \cdots$$

$$\langle\Psi_k^{(0)}(t)|i\hbar\partial_t - \hat{H}^{(0)}|\Psi_0^{(1)}(t)\rangle = \langle\Psi_k^{(0)}(t)|\hat{H}^{(1)}|\Psi_0^{(0)}(t)\rangle$$

$$(i\hbar\partial_t - \hat{H}^{(0)})\Psi_k^{(0)}(t) = 0$$

$$|\Psi_0^{(1)}(t)\rangle = \sum_{k\neq 0} a_k(t)|\Psi_k^{(0)}(t)\rangle$$

$$a_k(t) = \frac{1}{i\hbar}\int_0^t \langle\Psi_k^{(0)}(\tau)|\hat{H}^{(1)}(\tau)|\Psi_0^{(0)}(\tau)\rangle d\tau$$

代入 $\Psi_k^{(0)}(\boldsymbol{r},t) = \psi_k(\boldsymbol{r})e^{\frac{-iE_k^{(0)}t}{\hbar}}$ ➡ $a_k(t) = \frac{1}{i\hbar}\int_0^t \langle\psi_k^{(0)}|\hat{H}^{(1)}(\tau)|\psi_0^{(0)}\rangle e^{i\omega_{k0}\tau}d\tau$

引入激发态能量(频率)

$$\omega_{kn} = \frac{E_k^{(0)} - E_n^{(0)}}{\hbar}$$

$k\neq 0$ 可解读为跃迁概率幅

假设初态为基态

$$\hat{H}(t) = \hat{H}^{(0)} + \lambda \hat{H}^{(1)}(t)$$

$$|\Psi(t)\rangle = |\Psi_0^{(0)}(t)\rangle + \lambda|\Psi_0^{(1)}(t)\rangle + \lambda^2|\Psi_0^{(2)}(t)\rangle + \cdots$$

$$|\Psi_0^{(1)}(t)\rangle = \sum_n a_n(t)|\Psi_n^{(0)}(t)\rangle$$

利用了$\Psi_n^{(0)}(t)$的完备性

代入含时薛定谔方程

$$i\hbar\partial_t\Psi(\boldsymbol{r},t) = \hat{H}\Psi(\boldsymbol{r},t)$$

只考虑一阶微扰，得

$$i\hbar\partial_t|\Psi_0^{(1)}(t)\rangle = \hat{H}^{(0)}|\Psi_0^{(1)}(t)\rangle + \hat{H}^{(1)}(t)|\Psi_0^{(0)}(t)\rangle$$

一阶微扰方程与$\Psi_k^{(0)}(t)$做内积，得到

$$\langle\Psi_k^{(0)}(t)|i\hbar\partial_t|\Psi_0^{(1)}(t)\rangle = \langle\Psi_k^{(0)}(t)|\hat{H}^{(0)}|\Psi_0^{(1)}(t)\rangle + \langle\Psi_k^{(0)}(t)|\hat{H}^{(1)}(t)|\Psi_0^{(0)}(t)\rangle$$

代入$\Psi_0^{(1)}(t)$的展开式，并移项得

$$\sum_n\langle\Psi_k^{(0)}(t)|i\hbar\partial_t - \hat{H}^{(0)}|a_n(t)\Psi_n^{(0)}(t)\rangle = \langle\Psi_k^{(0)}(t)|\hat{H}^{(1)}(t)|\Psi_0^{(0)}(t)\rangle$$

$$\sum_n \langle \Psi_k^{(0)}(t)|i\hbar\partial_t - \hat{H}^{(0)}|a_n(t)\Psi_n^{(0)}(t)\rangle = \langle \Psi_k^{(0)}(t)|\hat{H}^{(1)}(t)|\Psi_0^{(0)}(t)\rangle$$

$$(i\hbar\partial_t - \hat{H}^{(0)})|a_n(t)\Psi_n^{(0)}(t)\rangle = \Big(i\hbar\partial_t a_n(t)\Big)|\Psi_n^{(0)}(t)\rangle + a_n(t)(i\hbar\partial_t - \hat{H}^{(0)})|\Psi_n^{(0)}(t)\rangle$$

利用定态性质 $(i\hbar\partial_t - \hat{H}^{(0)})|\Psi_n^{(0)}(t)\rangle = 0$

化简得

$$\sum_n i\hbar\partial_t a_n(t)\langle \Psi_k^{(0)}(t)|\Psi_n^{(0)}(t)\rangle = \langle \Psi_k^{(0)}(t)|\hat{H}^{(1)}(t)|\Psi_0^{(0)}(t)\rangle$$

引入激发态能量(频率)

$$\omega_{kn} = \frac{E_k^{(0)} - E_n^{(0)}}{\hbar}$$

计算内积

$$\langle \Psi_k^{(0)}(t)|\Psi_n^{(0)}(t)\rangle = \langle \psi_k^{(0)} e^{-\frac{iE_k^{(0)}t}{\hbar}}|\psi_n^{(0)} e^{-\frac{iE_n^{(0)}t}{\hbar}}\rangle = \delta_{kn} e^{\frac{i(E_k^{(0)}-E_n^{(0)})t}{\hbar}}$$

$$\langle \Psi_k^{(0)}(t)|\hat{H}^{(1)}(t)|\Psi_0^{(0)}(t)\rangle = \langle \psi_k^{(0)} e^{-\frac{iE_k^{(0)}t}{\hbar}}|\hat{H}^{(1)}(t)|\psi_0^{(0)} e^{-\frac{iE_0^{(0)}t}{\hbar}}\rangle = \langle \psi_k^{(0)}|\hat{H}^{(1)}(t)|\psi_0^{(0)}\rangle e^{\frac{i(E_k^{(0)}-E_0^{(0)})t}{\hbar}}$$

方程化简得

$$i\hbar\partial_t a_k(t) = \langle \psi_k^{(0)}|\hat{H}^{(1)}(t)|\psi_0^{(0)}\rangle e^{i\omega_{k0}t}$$

积分得一阶微扰展开系数

$$a_k(t) = \frac{1}{i\hbar}\int_0^t \langle \psi_k^{(0)}|\hat{H}^{(1)}(\tau)|\psi_0^{(0)}\rangle e^{i\omega_{k0}\tau} d\tau$$

$k\neq 0$ 可解读为跃迁概率幅

跃迁概率幅(transition probability amplitude)

$$a_k(t) = \frac{1}{i\hbar}\int_0^t \langle\psi_k^{(0)}|\hat{H}^{(1)}(\tau)|\psi_0^{(0)}\rangle e^{i\omega_{k0}\tau}d\tau$$

跃迁概率(transition probability) $P_k(t) = |a_k(t)|^2$

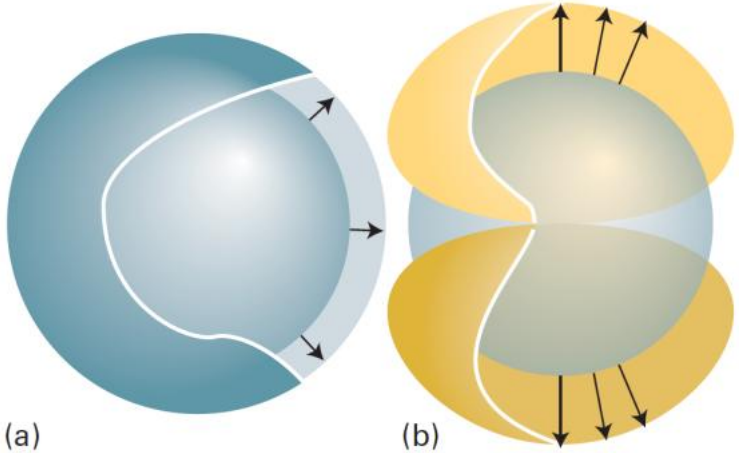
跃迁速率(transition rate) $w_k(t) = \frac{dP_k}{dt}$

光致跃迁：振荡的电场 $\hat{H}^{(1)}(t) = -\hat{\mu}_z\mathcal{E}\cos\omega t$

代入得

$$a_k(t) = -\mathcal{E}\langle\psi_k^{(0)}|\hat{\mu}_z|\psi_0^{(0)}\rangle\frac{1}{i\hbar}\int_0^t e^{i\omega_{k0}\tau}\cos\omega\tau d\tau$$

$$w_k(t) \propto \mathcal{E}^2|\langle\psi_k^{(0)}|\hat{\mu}_z|\psi_0^{(0)}\rangle|^2$$



1s→2s，禁阻　　1s → 2p，允许

跃迁概率/速率/**吸收(发射)强度**正比于偶极矩矩阵元的模方

$\mu_{k0} = \langle\psi_k^{(0)}|\hat{\mu}_z|\psi_0^{(0)}\rangle$　　跃迁偶极矩 (transition dipole moment)

非定态的能量-时间关系

考虑衰减的态 $\Psi(\boldsymbol{r},t)=\psi(\boldsymbol{r})e^{-iEt/\hbar}\times e^{-t/2\tau}$

τ 可理解为该态的**寿命(lifetime)**　　定态$\tau=\infty$

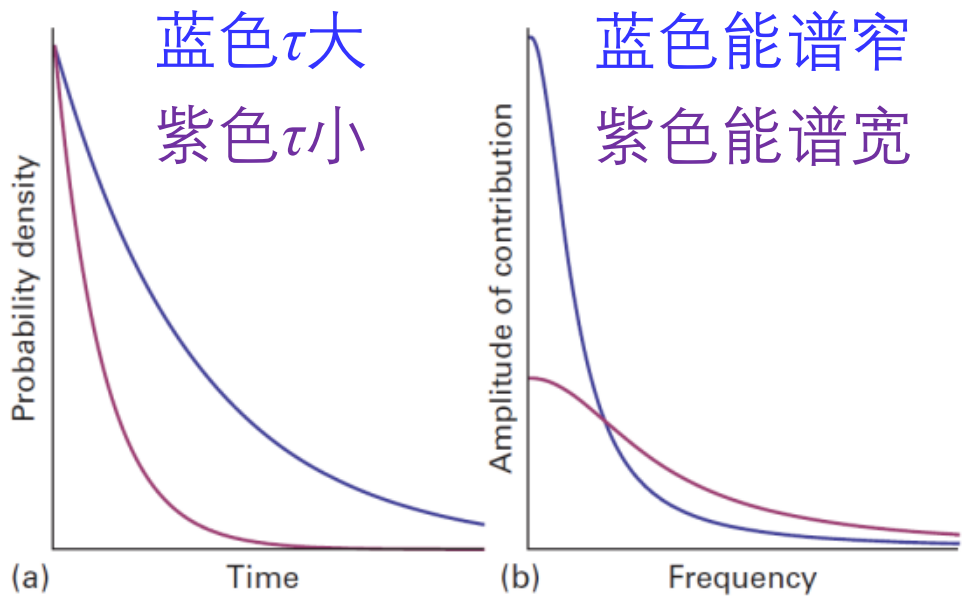
对能量本征态展开 $\Psi(\boldsymbol{r},t)=\psi(\boldsymbol{r})\sum_{\varepsilon}c_{\varepsilon}e^{-i(E+\varepsilon)t/\hbar}$

能量-时间不确定性关系

$$\Delta E\approx\frac{\hbar}{\tau}$$

$$\sigma_E\frac{\sigma_B}{\left|\frac{d\langle\hat{B}\rangle}{dt}\right|}\geq\frac{\hbar}{2}$$

$\hat{B}$是某可观测量，第二项相当于某状态的寿命



电磁场中的薛定谔方程——Pauli equation

$$i\hbar\partial_t\psi = \hat{H}\psi \qquad \hat{H} = \frac{1}{2m}\Big(\boldsymbol{\sigma}\cdot(\hat{\boldsymbol{p}} - e\boldsymbol{A})\Big)^2 + e\phi$$

先假定单电子

ϕ是标量势

$$\sigma_{\mathrm{x}} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_{\mathrm{y}} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_{\mathrm{z}} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

利用Pauli vector identity

$$(\boldsymbol{\sigma}\cdot\boldsymbol{a})(\boldsymbol{\sigma}\cdot\boldsymbol{b}) = \boldsymbol{a}\cdot\boldsymbol{b} + i\boldsymbol{\sigma}\cdot(\boldsymbol{a}\times\boldsymbol{b})$$

矢量微分恒等式

$$\nabla\times(\psi\boldsymbol{A}) = \nabla\psi\times\boldsymbol{A} + \psi\nabla\times\boldsymbol{A}$$

$$\Big[(\hat{\boldsymbol{p}} - e\boldsymbol{A})\times(\hat{\boldsymbol{p}} - e\boldsymbol{A})\Big]\psi = -e\hat{\boldsymbol{p}}\times(\boldsymbol{A}\psi) - e\boldsymbol{A}\times(\hat{\boldsymbol{p}}\psi)$$

$$= ie\hbar\Big[\nabla\times(\boldsymbol{A}\psi) + \boldsymbol{A}\times(\nabla\psi)\Big] = ie\hbar(\nabla\times\boldsymbol{A})\psi$$

哈密顿算符标准形式

$$\hat{H} = \frac{1}{2m}(\hat{\boldsymbol{p}} - e\boldsymbol{A})^2 + e\phi - \frac{e\hbar}{2m}\boldsymbol{\sigma}\cdot\boldsymbol{B} \qquad \boldsymbol{B} = \nabla\times\boldsymbol{A}$$

偶极近似（dipole approximation）

原子单位制 $\hbar = 1, \quad e = 1, \quad m_e = 1$

电磁场中的多电子哈密顿量

忽略原子核的量子性

$$\hat{H}(t) = \frac{1}{2}\sum_{j=1}^{N}\left[\frac{\nabla_j}{i} + \mathbf{A}(\mathbf{r}_j, t)\right]^2 + \sum_{j=1}^{N} v_0(\mathbf{r}_j) + \mu_B \sum_{j=1}^{N} \boldsymbol{\sigma}\cdot\nabla_j \times \mathbf{A}(\mathbf{r}_j, t)$$

$$+ \frac{1}{2}\sum_{\substack{j,k \\ j\neq k}}^{N} \frac{1}{|\mathbf{r}_j - \mathbf{r}_k|},$$

$\boldsymbol{A}(\boldsymbol{r}, t)$是矢量势

$\boldsymbol{\sigma}$是泡利矩阵，μ_B是玻尔磁子

上式成立条件：

- 电磁场被经典处理，要求光子数很多
- 非相对论处理，要求电子速度远小于光速

- 进一步假设电磁波长(几百nm)>>分子尺度(<1nm)，从而可以忽略$\boldsymbol{A}(\boldsymbol{r}, t)$随空间变化

$$\boldsymbol{A}(\boldsymbol{r}, t) \approx \boldsymbol{A}(t) \qquad \nabla \times \boldsymbol{A}(\boldsymbol{r}, t) = 0$$

如条件不成立，则需

- 量子电动力学——光场量子化
- 狄拉克方程——相对论量子力学

方程简化为

$$i\frac{\partial}{\partial t}\Psi(t)=\left(\frac{1}{2}\sum_{j=1}^{N}\left[\frac{\nabla_j}{i}+\mathbf{A}(t)\right]^2+\sum_{j=1}^{N}v_0(\mathbf{r}_j)+\frac{1}{2}\sum_{\substack{j,k\\ j\neq k}}^{N}\frac{1}{|\mathbf{r}_j-\mathbf{r}_k|}\right)\Psi(t)$$

作酉变换(规范变换)，将矢量势转化为标量式

$$\Psi'(t)=e^{i\phi(t)}\Psi(t)\qquad \phi(t)=\sum_{j=1}^{N}\mathbf{r}_j\cdot\mathbf{A}(t)$$

原方程变换为

$$i\frac{\partial}{\partial t}\Psi(t)=\hat{H}(t)\Psi(t)\quad\Longrightarrow\quad i\frac{\partial}{\partial t}\Psi'(t)=\hat{H}'(t)\Psi'(t)$$

其中

$$\hat{H}'(t)=e^{i\phi(t)}\hat{H}(t)e^{-i\phi(t)}-\frac{\partial\phi(t)}{\partial t}$$

最终薛定谔方程变为

$$i\frac{\partial}{\partial t}\Psi'(\mathbf{r},t)=\left(-\sum_{j=1}^{N}\frac{\nabla_j^2}{2}+\sum_{j=1}^{N}[\mathbf{r}_j\cdot\mathbf{E}(t)+v_0(\mathbf{r}_j)]+\frac{1}{2}\sum_{\substack{j,k\\ j\neq k}}^{N}\frac{1}{|\mathbf{r}_j-\mathbf{r}_k|}\right)\Psi'(t)$$

振荡电场

$$\mathbf{E}(t)=-\frac{\partial\mathbf{A}(t)}{\partial t}$$

复习：麦克斯韦方程与规范变换

一般形式的麦克斯韦方程

$$\begin{cases} \nabla \cdot \boldsymbol{E} = \dfrac{\rho}{\epsilon_0} \\ \nabla \cdot \boldsymbol{B} = 0 \\ \nabla \times \boldsymbol{E} = -\dfrac{\partial \boldsymbol{B}}{\partial t} \\ \nabla \times \boldsymbol{B} = \mu_0\left(\boldsymbol{J} + \epsilon_0 \dfrac{\partial \boldsymbol{E}}{\partial t}\right) \end{cases}$$

电场是有源场

磁场是无源场

ρ是电荷密度，$\boldsymbol{J}$是电流密度

ϵ_0是真空介电常数， μ_0是真空磁导率

真空中的麦克斯韦方程

$$\begin{cases} \nabla \cdot \boldsymbol{E} = 0 \\ \nabla \cdot \boldsymbol{B} = 0 \\ \nabla \times \boldsymbol{E} = -\dfrac{\partial \boldsymbol{B}}{\partial t} \\ \nabla \times \boldsymbol{B} = \mu_0\epsilon_0 \dfrac{\partial \boldsymbol{E}}{\partial t} \end{cases}$$

电磁辐射问题，选取**库伦规范** $\varphi = 0$

$$\nabla \cdot \boldsymbol{B} = 0$$

$$\nabla \times \boldsymbol{E} + \frac{\partial \boldsymbol{B}}{\partial t} = 0$$

$$\boldsymbol{B} = \nabla \times \boldsymbol{A}$$

$$\boldsymbol{E} = -\nabla\varphi - \frac{\partial \boldsymbol{A}}{\partial t}$$

$$\boldsymbol{B} = \nabla \times \boldsymbol{A}$$

$$\boldsymbol{E} = -\frac{\partial \boldsymbol{A}}{\partial t}$$

感生电场、无源磁场表述

规范不变量

标量式、矢量势表述

规范变量

也可取**洛伦兹规范** $\nabla \cdot \boldsymbol{A} = -\dfrac{1}{c^2}\dfrac{\partial \varphi}{\partial t}$

问题二　原子光谱

类氢原子

关键判断跃迁偶极矩是否为零 $\mu_{z,\mathrm{fi}} = -e\int \psi_\mathrm{f}^\star z \psi_\mathrm{i}\,\mathrm{d}\tau$

$$z = \sqrt{\frac{4\pi}{3}} r Y_{1,0}$$

$$y = i\sqrt{\frac{2\pi}{3}} r (Y_{1,1} + Y_{1,-1})$$

$$x = \sqrt{\frac{2\pi}{3}} r (Y_{1,-1} - Y_{1,1})$$

代入类氢原子轨道 $\int \psi_\mathrm{f}^\star z \psi_\mathrm{i}\,\mathrm{d}\tau =$

$$\int_0^\infty \int_0^{2\pi} \int_0^\pi \overbrace{R_{n_\mathrm{f},l_\mathrm{f}} Y^\star_{l_\mathrm{f},m_{l,\mathrm{f}}}}^{\psi_\mathrm{f}^\star} \overbrace{\left(\frac{4\pi}{3}\right)^{1/2} r Y_{1,0}}^{z} \overbrace{R_{n_\mathrm{i},l_\mathrm{i}} Y_{l_\mathrm{i},m_{l,\mathrm{i}}}}^{\psi_\mathrm{i}} \overbrace{r^2 \mathrm{d}r \sin\theta\, \mathrm{d}\theta \mathrm{d}\phi}^{\mathrm{d}\tau}$$

分离径向与角向积分

$$\int \psi_\mathrm{f}^\star z \psi_\mathrm{i}\,\mathrm{d}\tau =$$

$$\left(\frac{4\pi}{3}\right)^{1/2} \int_0^\infty R_{n_\mathrm{f},l_\mathrm{f}} r^3 R_{n_\mathrm{i},l_\mathrm{i}} \mathrm{d}r \int_0^{2\pi} \int_0^\pi Y^\star_{l_\mathrm{f},m_{l,\mathrm{f}}} Y_{1,0} Y_{l_\mathrm{i},m_{l,\mathrm{i}}} \sin\theta \mathrm{d}\theta \mathrm{d}\phi$$

定义一般的角向积分

$$I = \int_0^\pi \int_0^{2\pi} Y^\star_{l_\mathrm{f},m_{l,\mathrm{f}}} Y_{l,m} Y_{l_\mathrm{i},m_{l,\mathrm{i}}} \sin\theta \mathrm{d}\theta \mathrm{d}\phi$$

$I \neq 0$当且仅当 $l_\mathrm{f} = l_\mathrm{i} \pm l \quad m_{l,\mathrm{f}} = m_{l,\mathrm{i}} + m$

类氢原子跃迁选律

$\Delta l = \pm 1 \quad \Delta m_l = 0, \pm 1$

选律对主量子数没有要求

氦原子

双电子自旋组合

单重态 $\sigma_-(1,2) = \left(\frac{1}{2^{1/2}}\right)\{\alpha(1)\beta(2)-\beta(1)\alpha(2)\}$

三重态
$$\begin{cases} \alpha(1)\alpha(2) \\ \sigma_+(1,2) = \left(\frac{1}{2^{1/2}}\right)\{\alpha(1)\beta(2)+\beta(1)\alpha(2)\} \\ \beta(1)\beta(2) \end{cases}$$

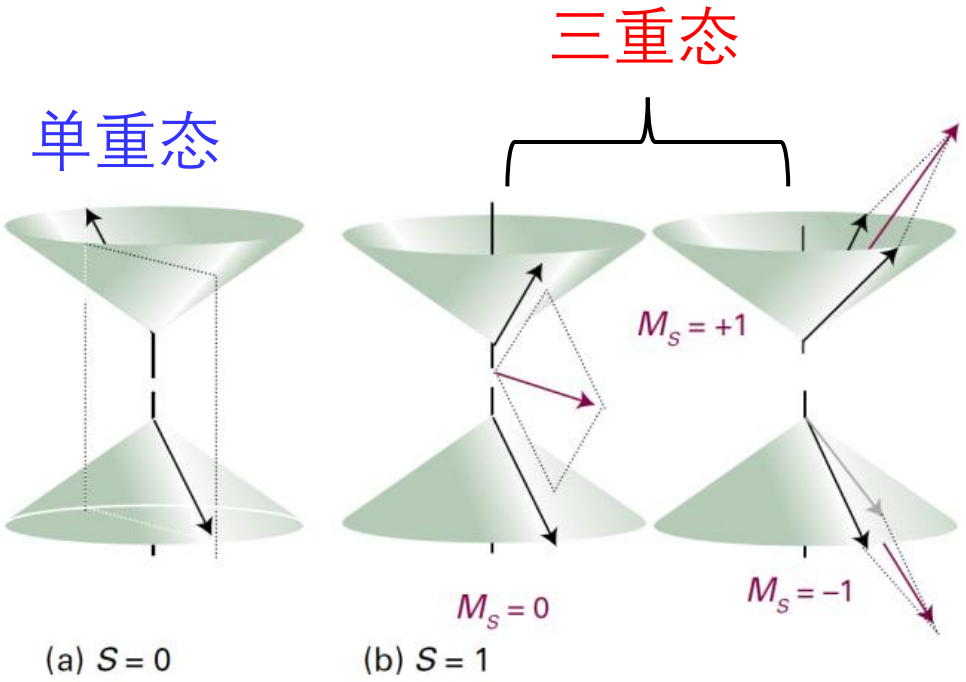
双电子自旋组合

基态$1s^2$，单重态

激发态$1s^1 2s^1$，$E_{triplet} < E_{singlet}$

高激发态$1s^1 nl^1$，$E_{triplet} < E_{singlet}$

复习洪特规则

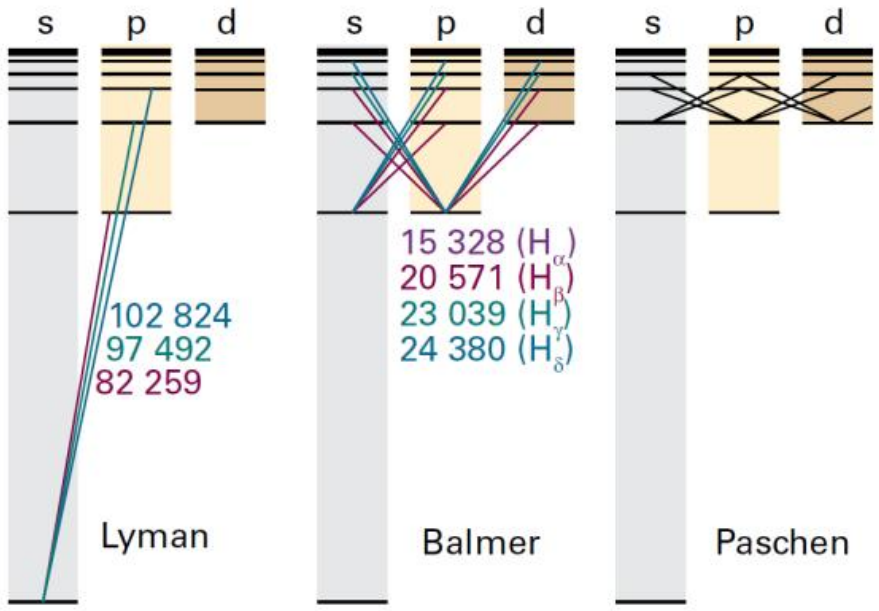


双电子自旋角动量的合成

复习：自旋量子数与自旋角动量的关系

He原子没有双激发态，因为激发能大于电离能

Grotrian diagram：原子光谱跃迁图

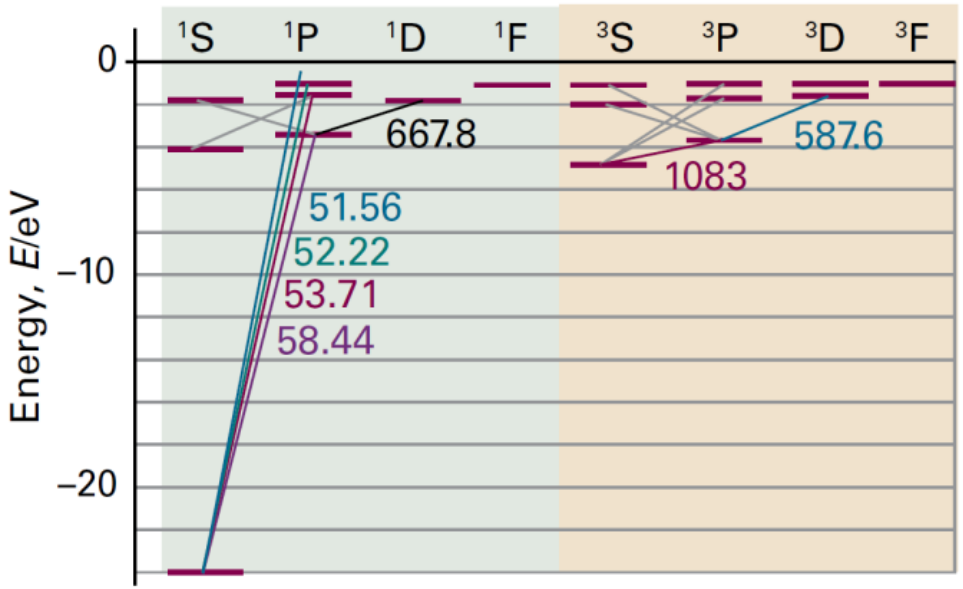


H原子

单电子原子用轨道符号表示各态

电子无法通过吸收光实现自旋翻转

单重态与三重态间无法跃迁！



He原子

多电子原子引入光谱项符号表示各态

真实的多电子原子没有轨道，只有波函数

自旋轨道耦合（spin-orbit coupling，SOC）

电子绕核旋转，产生**轨道角动量**，相当于环形电流，具有磁矩

电子自旋转，产生**自旋角动量**，也相当于环形电流，具有磁矩

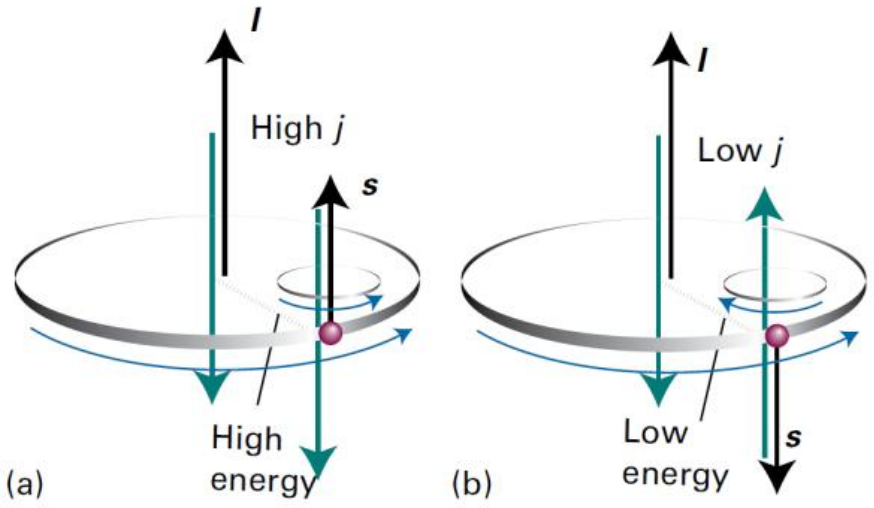
两种**磁矩相互作用**，倾向于**方向相反**

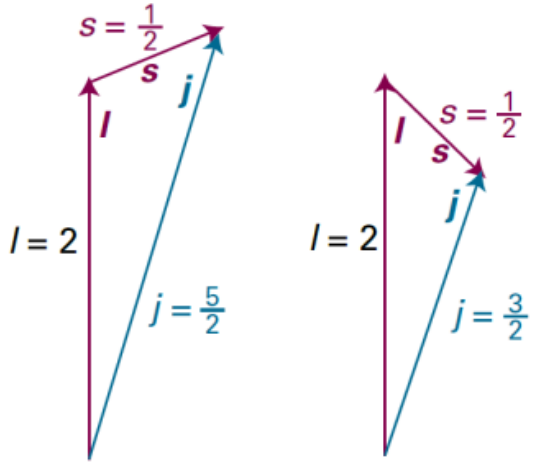
这一**相互作用**叫做**旋轨耦合**

总角动量：轨道角动量与自旋角动量的矢量加和

$$\hat{\mathcal{J}} = \hat{\mathcal{L}} + \hat{\mathcal{S}}$$

注意区别角动量与角动量量子数





用量子数表述的矢量叠加

单电子，j只能等于$l+s$或$l-s$

旋轨耦合能量表达式

旋轨耦合根本原因是相对论效应

经典力学中，磁矩在磁场中的能量为 $E = -\boldsymbol{\mu} \cdot \boldsymbol{B}$

磁场正比于轨道角动量，**磁矩**正比于自旋角动量

从而 $$E = -\boldsymbol{\mu} \cdot \boldsymbol{B} \propto \boldsymbol{s} \cdot \boldsymbol{l}$$

$$\overbrace{\boldsymbol{j}\cdot\boldsymbol{j}}^{j^2} = (\boldsymbol{l}+\boldsymbol{s})\cdot(\boldsymbol{l}+\boldsymbol{s}) = \overbrace{\boldsymbol{l}\cdot\boldsymbol{l}}^{l^2} + \overbrace{\boldsymbol{s}\cdot\boldsymbol{s}}^{s^2} + 2\boldsymbol{s}\cdot\boldsymbol{l} \quad \Rightarrow \quad j^2 = l^2 + s^2 + 2\boldsymbol{s}\cdot\boldsymbol{l}$$

$$\Rightarrow \quad \boldsymbol{s}\cdot\boldsymbol{l} = \tfrac{1}{2}\{j^2 - l^2 - s^2\}$$

将经典结论拓展到量子力学，利用角动量与量子数的关系

$$\boldsymbol{s}\cdot\boldsymbol{l} = \tfrac{1}{2}\{j(j+1) - l(l+1) - s(s+1)\}\hbar^2$$

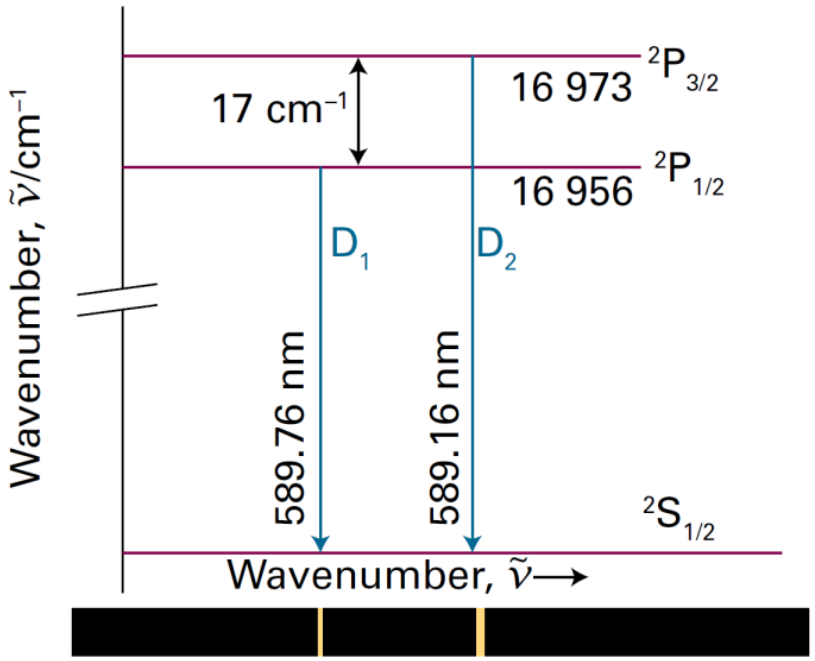
得**旋轨耦合能** $$E_{l,s,j} = \tfrac{1}{2}hc\tilde{A}\{j(j+1) - l(l+1) - s(s+1)\}$$

$\tilde{A}$是个波数，叫做旋轨耦合常数

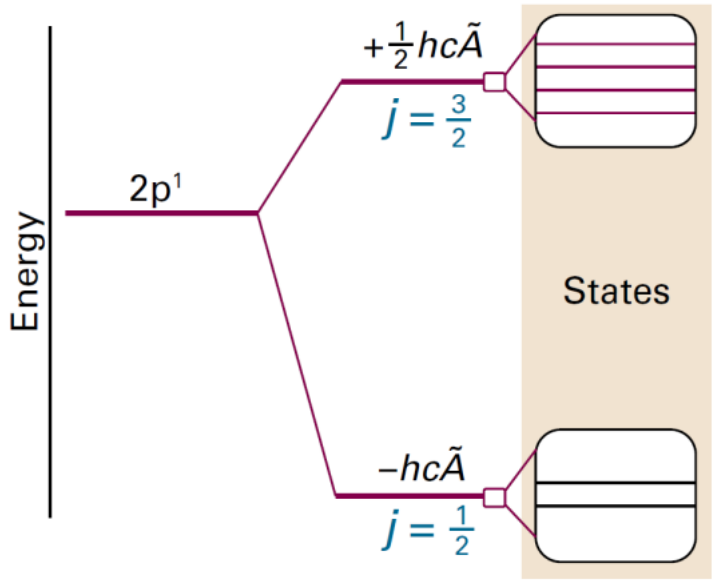
旋轨耦合大小与核电荷有关

$\propto Z^4$，从而重核SOC不可忽略

旋轨耦合导致光谱精细结构
(fine structure)



钠光灯光谱双线结构



悬轨耦合导致$2p^1$轨道的能级分裂

光谱项符号(term symbol)

$$^{2S+1}L_J$$

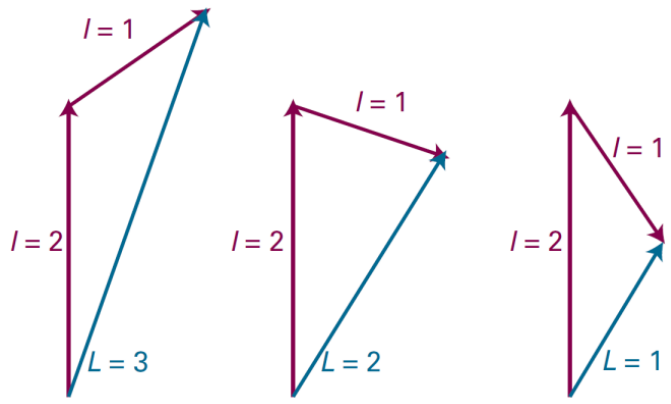
$2S+1$：自旋多重度

L：总轨道角动量量子数

J：总角动量量子数

总轨道角动量

以两个轨道为例



算符关系 $\hat{\boldsymbol{\mathcal{L}}} = \hat{\boldsymbol{\mathcal{L}}}_1 + \hat{\boldsymbol{\mathcal{L}}}_2$ 注意：矢量算符

"本征值" $|\mathcal{L}| = \sqrt{L(L+1)}\hbar$ $|\mathcal{L}_1| = \sqrt{l_1(l_1+1)}\hbar$ $|\mathcal{L}_2| = \sqrt{l_2(l_2+1)}\hbar$

这里默认指算符平方的本征值，或者开方得模长

量子数 L, l_1, l_2 注意：都是整数

总角动量量子数取值范围 $L = l_1 + l_2, l_1 + l_2 - 1, \ldots, |l_1 - l_2|$ Clebsch–Gordan series

模长满足三角不等式： $\Big||\mathcal{L}_1| - |\mathcal{L}_2|\Big| \leqslant |\mathcal{L}| \leqslant |\mathcal{L}_1| + |\mathcal{L}_2|$

角动量z分量 $|\mathcal{L}_z| = m_z\hbar$ $|\mathcal{L}_{1z}| = m_{1z}\hbar$ $|\mathcal{L}_{2z}| = m_{2z}\hbar$

分量量子数 m_z, m_{1z}, m_{2z}

总角动量分量量子数取值范围 $m_z = 0, \pm 1, \pm 2, \cdots, \pm L$

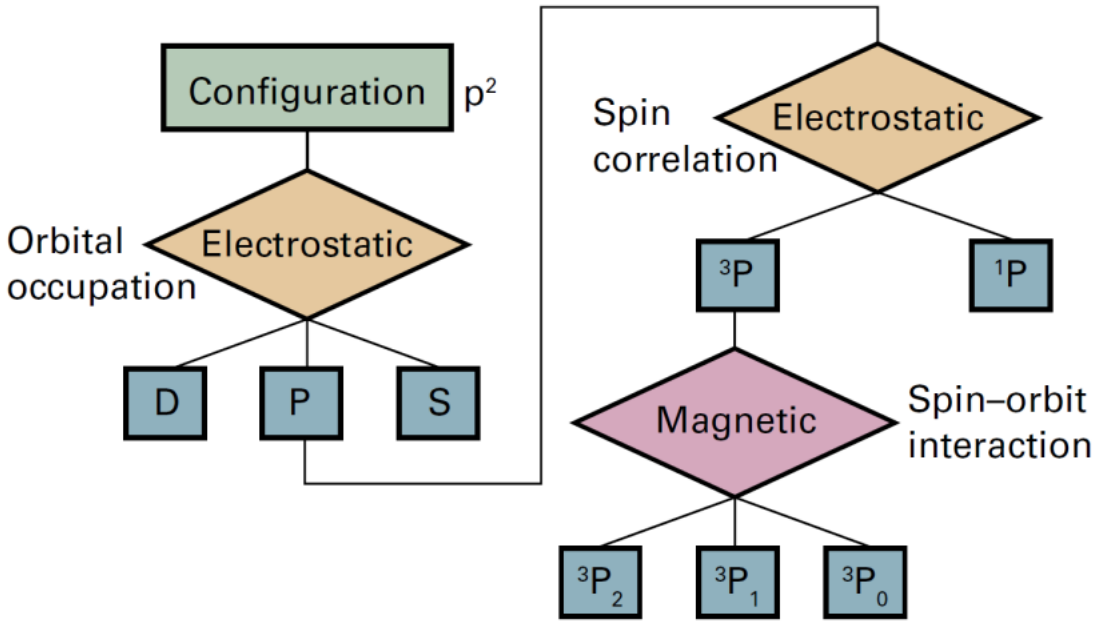
总轨道角动量符号对应

L:	0	1	2	3	4	5	6…
	S	P	D	F	G	H	I…

注意：都是正体字

闭壳层L=0　思考为什么？

从而[Ne]$3s^1$光谱项为S



导致原子能级分裂的因素

例题：(a) d^2有哪些光谱项？

(a) d^2　$L = 2+2, 2+2-1, \ldots, 0 = 4, 3, 2, 1, 0$

对应G, F, D, P, S

每个电子可能状态6种

2个电子可能状态$C_6^2 = 15$种

5个 ^1D

9个 ^3P

1个 ^1S

思考:
$(2p)^1(3p)^1$组态可能的微观状态数以及光谱项

表 2.6.4 $(np)^2$ 组态的 15 种微观状态

m			$m_L = \sum m_l$	$m_S = \sum m_s$	T_n
1	0	−1			
↑↓			2	0	^1D
↑	↑		1	1	^3P
↑	↓		1	0	^1D, ^3P
↓	↑		1	0	
↑		↑	0	1	^3P
↑		↓	0	0	^1D, ^3P, ^1S
↓		↑	0	0	
	↑↓		0	0	
↓	↓		1	−1	^3P
↓		↓	0	−1	^3P
	↑	↑	−1	1	^3P
	↑	↓	−1	0	^1D, ^3P
	↓	↑	−1	0	
	↓	↓	−1	−1	^3P
		↑↓	−2	0	^1D

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	m	
1	0	-1
↑	↓	
↓	↑	

为什么组合不出 ^1P？

$\Rightarrow$ ^3P, ^1D

$$\Phi_\pm(\boldsymbol{x}_1,\boldsymbol{x}_2) = N\Big(\psi_{+1}(\boldsymbol{r}_1)\psi_0(\boldsymbol{r}_2) \pm \psi_{+1}(\boldsymbol{r}_2)\psi_0(\boldsymbol{r}_1)\Big)\Big(\alpha(1)\beta(2) \mp \alpha(2)\beta(1)\Big)$$

$$= N\Psi_\pm(\boldsymbol{r}_1,\boldsymbol{r}_2)\chi_\mp(1,2)$$

谁是单重态，谁是三重态?

$$\hat{\boldsymbol{L}}^2\Psi_\pm(\boldsymbol{r}_1,\boldsymbol{r}_2) = (\hat{\boldsymbol{L}}_1+\hat{\boldsymbol{L}}_2)^2\Psi_\pm(\boldsymbol{r}_1,\boldsymbol{r}_2) = (\hat{\boldsymbol{L}}_1^2+\hat{\boldsymbol{L}}_2^2+2\hat{\boldsymbol{L}}_1\cdot\hat{\boldsymbol{L}}_2)\Psi_\pm(\boldsymbol{r}_1,\boldsymbol{r}_2)$$

$$\hat{\boldsymbol{L}}_1^2\Psi_\pm(\boldsymbol{r}_1,\boldsymbol{r}_2) = 1\times(1+1)\hbar^2\Psi_\pm(\boldsymbol{r}_1,\boldsymbol{r}_2) = \hat{\boldsymbol{L}}_2^2\Psi_\pm(\boldsymbol{r}_1,\boldsymbol{r}_2)$$

$$\hat{\boldsymbol{L}}_1\cdot\hat{\boldsymbol{L}}_2\Psi_\pm(\boldsymbol{r}_1,\boldsymbol{r}_2) = \pm\hbar^2\Psi_\pm(\boldsymbol{r}_1,\boldsymbol{r}_2) \Longrightarrow \begin{cases} \hat{\boldsymbol{L}}^2\Psi_+(\boldsymbol{r}_1,\boldsymbol{r}_2) = 6\hbar^2\Psi_+(\boldsymbol{r}_1,\boldsymbol{r}_2) & \Longrightarrow \quad L=2 \\ \hat{\boldsymbol{L}}^2\Psi_-(\boldsymbol{r}_1,\boldsymbol{r}_2) = 2\hbar^2\Psi_-(\boldsymbol{r}_1,\boldsymbol{r}_2) & \Longrightarrow \quad L=1 \end{cases}$$

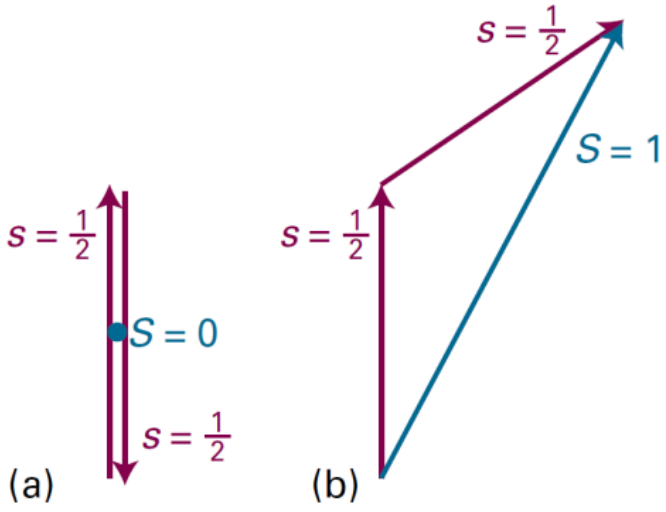
总自旋角动量

总轨道角动量类似，以两电子为例

算符关系，本征值(模长)，量子数

$$S = s_1 + s_2, s_1 + s_2 - 1, \ldots, |s_1 - s_2| \qquad s_i = \frac{1}{2}$$

总自旋角动量量子数取值范围，z分量量子数以及取值范围



自旋配对　　自旋未配对

[Ne]$3s^1$为二重态，光谱项为^{2}S

总角动量

单电子时，轨道与自旋耦合

$J = j$，取值范围为$l + \frac{1}{2}$或$|l - \frac{1}{2}|$

例如 [Ne]$3s^1$，$j = \frac{1}{2}$，光谱项为$^2S_{1/2}$

[Ne]$3p^1$，$j = \frac{3}{2}$或$\frac{1}{2}$，光谱项为$^2P_{3/2}$或$^2P_{1/2}$

Na原子双线结构，旋轨耦合导致分裂

多电子总角动量：两种耦合方式

1. 总轨道角动量L与总自旋角动量S耦合 (Russell–Saunders coupling)

适用于弱旋轨耦合(轻核)

总角动量量子数可取值： $J = L + S, L + S - 1, \ldots, |L - S|$

例题：写出(a) Na, (b) F基态构型，以及(c) C的激发态 $1s^2 2s^2 2p^1 3p^1$的光谱项

解答：(a) Na基态构型[Ne]$3s^1$，光谱项为$^2S_{1/2}$

(b) F基态构型[He]$2s^2 2p^5$ = [Ne]$2p^{-1}$， $L = 1, S = \frac{1}{2}$光谱项为$^2P_{3/2}$ 或$^2P_{1/2}$

(c) C激发态构型$1s^2 2s^2 2p^1 3p^1$， 两电子问题

$l_1 = l_2 = 1$

$s_1 = s_2 = \frac{1}{2}$

→ $L = 0,1,2$

$S = 0,1$

→

$^1D \rightarrow {}^1D_2$

$^1P \rightarrow {}^1P_1$

$^1S \rightarrow {}^1S_0$

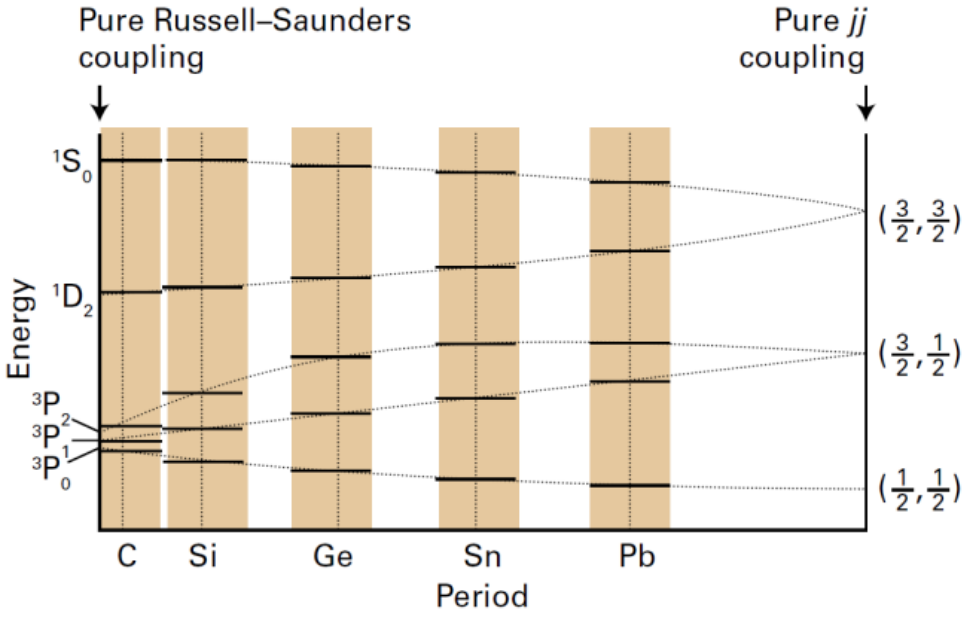
$^3D \rightarrow {}^3D_3, {}^3D_2, {}^3D_1$

$^3P \rightarrow {}^3P_2, {}^3P_1, {}^3P_0$

$^3S \rightarrow {}^3S_1$

2. 每个电子轨道角动量l与自旋角动量s耦合成 j,
然后不同电子再耦合(jj-coupling)

适用于强旋轨耦合(重核)



RS耦合与jj耦合图像之间的关联

各能态真实能量介于两种图像之间

洪特(Hund)规则：

- **给定构型，多重度最大的光谱项能量最低**
- **给定S，L最大的光谱项能量最低**

  经典解读：电子倾向于同向旋转（相遇机率低）
- **小于半满，J最小能量最低**
  大于半满，J最大能量最低

跃迁选律：

$$\Delta S = 0,$$

$$\Delta L = 0, \pm 1, \Delta l = \pm 1,$$

$$\Delta J = 0, \pm 1 \text{ but } J = 0 \not\longleftrightarrow J = 0$$

适用RS耦合，有反例

洪特规则的进一步说明：

$$\Delta E = \zeta(L,S)\{\mathbf{L}\cdot\mathbf{S}\}$$
$$= (1/2)\zeta(L,S)\{J(J+1) - L(L+1) - S(S+1)\}$$

- 小于半满，$\zeta(L,S) > 0$，J 小能量低
- 大于半满，$\zeta(L,S) < 0$，J 大能量低

洪特规则的应用：

$$\text{Configurations} \xrightarrow[\text{repulsions}]{\text{interelectronic}} \text{Terms} \xrightarrow[\text{interaction}]{\text{spin–orbit}} \text{Levels} \xrightarrow[\text{field}]{\text{external}} \text{States}$$

- Si原子：$(2p)^2$，光谱项 ^1D，^1S，^3P（基态） 基态支项是 $^3P_0$
- Ti原子：$(3d)^2$，光谱项 ^1S，^1D，^1G，^3P，^3F（基态）
- S原子基态支项是 $^3P_2$

思考：基态支项

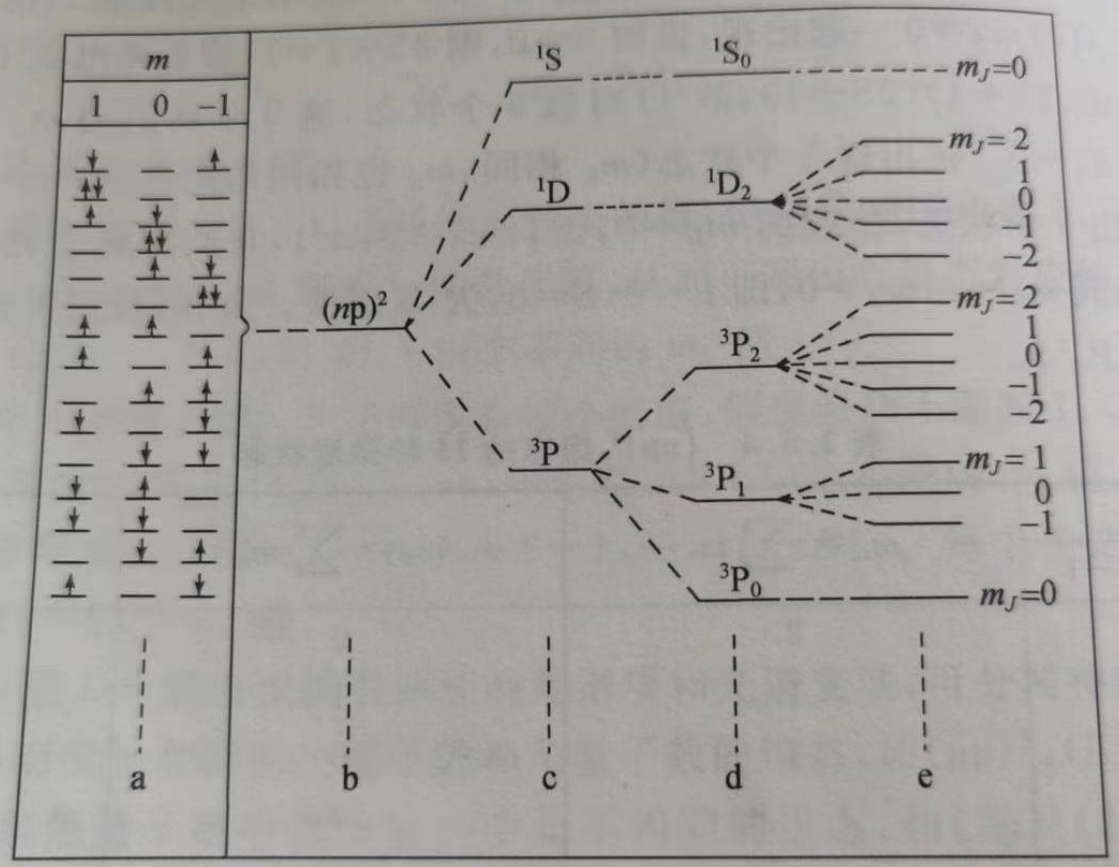


图 2.6.4 $(np)^2$ 组态的能级分裂

a—微状态(有磁场);b—组态,不考虑电子相互作用(无磁场);
c—谱项,分别考虑电子的轨道和自旋的作用(无磁场);
d—光谱支项,考虑 L-S 的相互作用(无磁场);e—微能态(有磁场)

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应用洪特规则解释$(np)^2$各组态的能级顺序

能级在外磁场中的分裂称为塞曼（Zeeman）效应

补充问题　旋轨耦合算符的表示

有限维向量的表示

(1) $\boldsymbol{u} = (u_1, u_2, \cdots, u_N)$

(2) $\boldsymbol{u} = u_1\boldsymbol{e}_1 + u_2\boldsymbol{e}_2 + \cdots + u_N\boldsymbol{e}_N = (u_1, u_2, \cdots, u_N)\begin{pmatrix} \boldsymbol{e}_1 \\ \boldsymbol{e}_2 \\ \cdots \\ \boldsymbol{e}_N \end{pmatrix}$

(3) $\langle \boldsymbol{e}_k | \boldsymbol{u} \rangle = u_k$

无限维函数的表示

一般用第(3)种 $\langle x|\psi\rangle = \psi(x)$

$$\left(\psi_1(x), \psi_2(x), \cdots, \psi_N(x)\right)\begin{pmatrix} |\boldsymbol{e}_1\rangle \\ |\boldsymbol{e}_2\rangle \\ \cdots \\ |\boldsymbol{e}_N\rangle \end{pmatrix}$$

既含离散变量，又含连续变量的“函数”的表示

$$\langle x, \boldsymbol{e}_k|\psi\rangle = \psi_k(x) \quad \Longleftrightarrow \quad |\psi\rangle = \psi_1(x)|\boldsymbol{e}_1\rangle + \psi_2(x)|\boldsymbol{e}_2\rangle + \cdots + \psi_N(x)|\boldsymbol{e}_N\rangle$$

$\psi_k(x)$ 有时也写成 $\psi(x, k)$

自旋波函数的写法

单电子：

$$|\chi\rangle = c_\alpha|\alpha\rangle + c_\beta|\beta\rangle \qquad \chi(1) = c_\alpha\alpha(1) + c_\beta\beta(1)$$

$$\begin{cases} \hat{S}_z|\alpha\rangle = \frac{\hbar}{2}|\alpha\rangle \\ \hat{S}_z|\beta\rangle = -\frac{\hbar}{2}|\beta\rangle \end{cases} \Longrightarrow \quad \hat{S}_z\begin{pmatrix}|\alpha\rangle \\ |\beta\rangle\end{pmatrix} = \frac{\hbar}{2}\begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}\begin{pmatrix}|\alpha\rangle \\ |\beta\rangle\end{pmatrix} = \frac{\hbar}{2}\boldsymbol{\sigma}_z\begin{pmatrix}|\alpha\rangle \\ |\beta\rangle\end{pmatrix}$$

$$\sigma_{\mathrm{x}} = \begin{pmatrix}0 & 1 \\ 1 & 0\end{pmatrix} \quad \sigma_{\mathrm{y}} = \begin{pmatrix}0 & -i \\ i & 0\end{pmatrix} \quad \sigma_{\mathrm{z}} = \begin{pmatrix}1 & 0 \\ 0 & -1\end{pmatrix}$$

$\hat{S}_z$的表示矩阵为$\frac{\hbar}{2}\sigma_z$

$$\begin{cases} \hat{\boldsymbol{S}}^2|\alpha\rangle = \frac{3}{4}\hbar^2|\alpha\rangle \\ \hat{\boldsymbol{S}}^2|\beta\rangle = \frac{3}{4}\hbar^2|\beta\rangle \end{cases} \quad \hat{\boldsymbol{S}}^2\begin{pmatrix}|\alpha\rangle \\ |\beta\rangle\end{pmatrix} = \frac{3\hbar^2}{4}\begin{pmatrix}1 & 0 \\ 0 & 1\end{pmatrix}\begin{pmatrix}|\alpha\rangle \\ |\beta\rangle\end{pmatrix} = \frac{\hbar^2}{4}(\boldsymbol{\sigma}_x^2 + \boldsymbol{\sigma}_y^2 + \boldsymbol{\sigma}_z^2)\begin{pmatrix}|\alpha\rangle \\ |\beta\rangle\end{pmatrix}$$

$\hat{\boldsymbol{S}}^2$的表示矩阵为$\frac{\hbar^2}{4}(\sigma_x^2+\sigma_y^2+\sigma_z^2)$

自旋波函数的写法

双电子：

$$|\chi\rangle = c_{\alpha\alpha}|\alpha\alpha\rangle + c_{\alpha\beta}|\alpha\beta\rangle + c_{\beta\alpha}|\beta\alpha\rangle + c_{\beta\beta}|\beta\beta\rangle$$

$$\chi(1,2) = c_{\alpha\alpha}\alpha(1)\alpha(2) + c_{\alpha\beta}\alpha(1)\beta(2) + c_{\beta\alpha}\beta(1)\alpha(2) + c_{\beta\beta}\beta(1)\beta(2)$$

$$\begin{cases} \left(\hat{S}_z(1)+\hat{S}_z(2)\right)\alpha(1)\alpha(2) = \hbar\alpha(1)\alpha(2) \\ \left(\hat{S}_z(1)+\hat{S}_z(2)\right)\alpha(1)\beta(2) = 0 \\ \left(\hat{S}_z(1)+\hat{S}_z(2)\right)\beta(1)\alpha(2) = 0 \\ \left(\hat{S}_z(1)+\hat{S}_z(2)\right)\beta(1)\beta(2) = -\hbar\beta(1)\beta(2) \end{cases} \Rightarrow \left(\hat{S}_z(1)+\hat{S}_z(2)\right)\begin{pmatrix}\alpha(1)\alpha(2)\\ \alpha(1)\beta(2)\\ \beta(1)\alpha(2)\\ \beta(1)\beta(2)\end{pmatrix} = \hbar\begin{pmatrix}1&0&0&0\\0&0&0&0\\0&0&0&0\\0&0&0&-1\end{pmatrix}\begin{pmatrix}\alpha(1)\alpha(2)\\ \alpha(1)\beta(2)\\ \beta(1)\alpha(2)\\ \beta(1)\beta(2)\end{pmatrix}$$

思考：总自旋角动量的平方 $\hat{\boldsymbol{S}}^2 = (\hat{\boldsymbol{S}}(\mathbf{1}) + \hat{\boldsymbol{S}}(\mathbf{2}))^2$
在这组基下的表示矩阵是什么？是对角矩阵吗？

算符$\hat{S}_z = \hat{S}_z(1) + \hat{S}_z(2)$
的表示矩阵

$$\left(\hat{S}_z(1)+\hat{S}_z(2)\right)\chi(1,2)=\left(\hat{S}_z(1)+\hat{S}_z(2)\right)\left(c_{\alpha\alpha}\alpha(1)\alpha(2)+c_{\alpha\beta}\alpha(1)\beta(2)+c_{\beta\alpha}\beta(1)\alpha(2)+c_{\beta\beta}\beta(1)\beta(2)\right)$$

$$=\hbar\left(c_{\alpha\alpha}\alpha(1)\alpha(2)-c_{\beta\beta}\beta(1)\beta(2)\right)$$

$$=\left(\alpha(1)\alpha(2),\alpha(1)\beta(2),\beta(1)\alpha(2),\beta(1)\beta(2)\right)\hbar\begin{pmatrix}c_{\alpha\alpha}\\0\\0\\-c_{\beta\beta}\end{pmatrix}$$

左边算符与波函数的作用结果在以上基组下表示为列向量

作用之前，$\chi(1,2)$在以上基组下也表示为列向量

基组表示下的作用方程 ➡ $$\boldsymbol{S}_z\begin{pmatrix}c_{\alpha\alpha}\\c_{\alpha\beta}\\c_{\beta\alpha}\\c_{\beta\beta}\end{pmatrix}=\hbar\begin{pmatrix}c_{\alpha\alpha}\\0\\0\\-c_{\beta\beta}\end{pmatrix}$$

空间-自旋全波函数的写法：以两电子为例

$$\Psi(\boldsymbol{r}_1, s_1, \boldsymbol{r}_2, s_2) = c_{\alpha\alpha}\Psi(\boldsymbol{r}_1, \alpha, \boldsymbol{r}_2, \alpha) + c_{\alpha\beta}\Psi(\boldsymbol{r}_1, \alpha, \boldsymbol{r}_2, \beta)$$

$$+ c_{\beta\alpha}\Psi(\boldsymbol{r}_1, \beta, \boldsymbol{r}_2, \alpha) + c_{\beta\beta}\Psi(\boldsymbol{r}_1, \beta, \boldsymbol{r}_2, \beta)$$

$$= c_{\alpha\alpha}\Psi_{\alpha\alpha}(\boldsymbol{r}_1, \boldsymbol{r}_2)\alpha(1)\alpha(2) + c_{\alpha\beta}\Psi_{\alpha\beta}(\boldsymbol{r}_1, \boldsymbol{r}_2)\alpha(1)\beta(2)$$

$$+ c_{\beta\alpha}\Psi_{\beta\alpha}(\boldsymbol{r}_1, \boldsymbol{r}_2)\beta(1)\alpha(2) + c_{\beta\beta}\Psi_{\beta\beta}(\boldsymbol{r}_1, \boldsymbol{r}_2)\beta(1)\beta(2)$$

$$\Psi(\boldsymbol{r}_1, s_1, \boldsymbol{r}_2, s_2) \longrightarrow \begin{pmatrix} c_{\alpha\alpha}\Psi_{\alpha\alpha}(\boldsymbol{r}_1, \boldsymbol{r}_2) \\ c_{\alpha\beta}\Psi_{\alpha\beta}(\boldsymbol{r}_1, \boldsymbol{r}_2) \\ c_{\beta\alpha}\Psi_{\beta\alpha}(\boldsymbol{r}_1, \boldsymbol{r}_2) \\ c_{\beta\beta}\Psi_{\beta\beta}(\boldsymbol{r}_1, \boldsymbol{r}_2) \end{pmatrix}$$

$$\longrightarrow \psi(\boldsymbol{r}_1, \boldsymbol{r}_2) \begin{pmatrix} c_{\alpha\alpha} \\ c_{\alpha\beta} \\ c_{\beta\alpha} \\ c_{\beta\beta} \end{pmatrix}$$

如果空间波函数相同，则

$$\Psi(\boldsymbol{r}_1, s_1, \boldsymbol{r}_2, s_2) = \psi(\boldsymbol{r}_1, \boldsymbol{r}_2)\Big(c_{\alpha\alpha}\alpha(1)\alpha(2) + c_{\alpha\beta}\alpha(1)\beta(2)$$

$$+ c_{\beta\alpha}\beta(1)\alpha(2) + c_{\beta\beta}\beta(1)\beta(2)\Big)$$

$$= \psi(\boldsymbol{r}_1, \boldsymbol{r}_2)\chi(1, 2)$$

$$\hat{S}_z\Psi(\boldsymbol{r}_1, s_1, \boldsymbol{r}_2, s_2) = (\hat{S}_z^1 + \hat{S}_z^2)\Psi(\boldsymbol{r}_1, s_1, \boldsymbol{r}_2, s_2) \longrightarrow \hbar \begin{pmatrix} c_{\alpha\alpha}\Psi_{\alpha\alpha}(\boldsymbol{r}_1, \boldsymbol{r}_2) \\ 0 \\ 0 \\ -c_{\beta\beta}\Psi_{\beta\beta}(\boldsymbol{r}_1, \boldsymbol{r}_2) \end{pmatrix}$$

$$\begin{aligned}
\hat{L}_z\Psi(\boldsymbol{r}_1, s_1, \boldsymbol{r}_2, s_2) &= (\hat{L}_z^1 + \hat{L}_z^2)\Psi(\boldsymbol{r}_1, s_1, \boldsymbol{r}_2, s_2) \\
&= (x_1\hat{p}_{y_1} - y_1\hat{p}_{x_1} + x_2\hat{p}_{y_2} - y_2\hat{p}_{x_2})\Psi(\boldsymbol{r}_1, s_1, \boldsymbol{r}_2, s_2) \\
&= -i\hbar(x_1\partial_{y_1} - y_1\partial_{x_1} + x_2\partial_{y_2} - y_2\partial_{x_2})\Psi(\boldsymbol{r}_1, s_1, \boldsymbol{r}_2, s_2) \\
&\longrightarrow -i\hbar(x_1\partial_{y_1} - y_1\partial_{x_1} + x_2\partial_{y_2} - y_2\partial_{x_2}) \begin{pmatrix} c_{\alpha\alpha}\Psi_{\alpha\alpha}(\boldsymbol{r}_1, \boldsymbol{r}_2) \\ c_{\alpha\beta}\Psi_{\alpha\beta}(\boldsymbol{r}_1, \boldsymbol{r}_2) \\ c_{\beta\alpha}\Psi_{\beta\alpha}(\boldsymbol{r}_1, \boldsymbol{r}_2) \\ c_{\beta\beta}\Psi_{\beta\beta}(\boldsymbol{r}_1, \boldsymbol{r}_2) \end{pmatrix}
\end{aligned}$$